Yousef Kart

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Experience

Red Piston — April 2026 – Present
Full Stack Software Developer — *Windsor, Ontario*

- Built and deployed an HR platform using Next.js, Supabase, and Vercel, implementing authentication, AI-powered employee support, applicant tracking, lead automation, and PTO scheduling
- Rebuilt a legacy Android application using React Native (Expo) and Strapi, expanding support to iOS, and web
- Improved existing products through testing, debugging, and code reviews, delivering production-ready fixes and enhancements across multiple applications

Scelta — February 2025 – September 2025
Software Engineer — *Windsor, Ontario*

- Led the design and rollout of a shared UI component library used across several production apps, reducing duplicated UI code and accelerating feature delivery
- Built an SSO system enabling seamless authentication across 3 applications, eliminating redundant login flows and improving session security
- Developed a real-time 3D geospatial visualization pipeline for interactive development mapping tools
- Reduced REDMap's initial loading time by 97% via query optimization and endpoint restructuring
- Designed a production-ready Next.js application template that decreased new project build time by more than 50%

Freelance — June 2021 – February 2025
3D Modeler — *Remote*

- Delivered 3D assets and rendering pipelines for 40+ global clients, optimizing polygon counts and texture workflows to meet real-time performance constraints
- Produced architectural visualizations for $100M+ developments while reducing asset turnaround times to 3–5 days

Skills

Languages: C++, Python, Java, TypeScript, SQL
Graphics & Interfaces: OpenGL, ImGui, Qt, WebRTC, Three.js, Mapbox
Frontend: React, React Native, Next.js, Expo, Tailwind, shadcn/ui
Backend & Database: Node.js, Supabase, Neon, Strapi, PostgreSQL, SQLite, REST APIs
Tools: CMake, Git, Vercel, OpenCV

Projects

Monte Carlo Pricing Engine | *C++20, OpenGL, ImGui*

- Built a high-performance Monte Carlo engine to simulate thousands of stochastic asset price paths using GBM
- Developed a real-time OpenGL renderer to visualize path dispersion and terminal payoff distributions
- Implemented interactive ImGui control panel enabling dynamic volatility, drift, timestep, and simulation scaling

Black-Scholes Model Visualizer | *C++17, Qt*

- Implemented Black-Scholes' pricing engine with real-time Greeks visualization and surface rendering
- Computed option sensitivities (Delta, Gamma, Vega, Theta, Rho) with interactive parameter controls
- Optimized computation pipeline to handle millions of floating-point operations per frame

Eyez and Earz | *Python, SQLite, LLMs, Finnhub, Massive, SEC EDGAR, yfinance*

- Built an event-driven trading pipeline that ingests real-time financial news and classifies market-moving events
- Engineered a modular data pipeline to process live news feeds and track market snapshots
- Developed a scoring and signal framework combining sentiment analysis, event classification, liquidity metrics, relative volume, and volatility detection to rank high-probability trading setups and reduce false-positives

Education

University of Windsor | *4x Dean's Honour Roll* — September 2021 – May 2025
Bachelor of Computer Science (Honours) & Minor in Mathematics — *Windsor, Ontario*